

Task 8 – Executive Business Report

Retail Sales Forecasting & Demand Segmentation

Prepared For: Head of Supply Chain & Chief Financial Officer (CFO)

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1. Executive Summary

This report reviews historical retail sales to support better inventory planning and business decisions. The analysis examined sales trends, identified unusual sales periods, estimated demand for the next three months and grouped products according to demand behaviour. Among the forecasting approaches evaluated, **XGBoost** provided the most reliable results. The findings indicate opportunities to improve stock availability, reduce excess inventory and strengthen demand planning across the business.

2. Key Findings

- Sales followed an overall positive trend with predictable seasonal fluctuations.
- Product demand was not uniform; some categories consistently outperformed others.
- A limited number of unusual sales periods were identified and deserve management attention.
- The selected forecasting model provides a dependable basis for short-term planning.

3. Three-Month Sales Forecast

Period	Expected Sales
Month 1	86,441
Month 2	91,658
Month 3	89,413

The forecast suggests that sales are expected to remain strong over the coming three months. Month 2 is projected to record the highest sales, followed by a slight moderation in Month 3. These projections should be used as planning guidance and reviewed regularly as new sales information becomes available.

4. Top Three Unusual Sales Events

1. A significant increase in sales was likely associated with promotional campaigns or festive shopping.
2. Another unusually high sales period may have resulted from bulk customer purchases.
3. A temporary decline in sales may indicate supply constraints, seasonal slowdown or reduced customer demand.

5. Product Demand Segmentation

Demand Segment	Recommended Stocking Strategy
High Value, Rapid Growth	Maintain higher inventory and prioritise replenishment.
Growing Demand	Increase stock gradually to meet expected demand.

High Volume, High Demand	Maintain safety stock and replenish frequently.
Low Volume, Stable Demand	Avoid overstocking and review inventory periodically.

6. Business Recommendations

Recommendation 1: Adopt **XGBoost** for monthly sales planning because it achieved the best forecasting performance.

Recommendation 2: Increase inventory for products in high-growth demand segments to minimise stock shortages during peak demand.

Recommendation 3: Review unusual sales periods before making procurement decisions, as these events may represent exceptional business activity rather than normal demand.

7. Risk / Limitation

Sales forecasts are based on historical business performance. Unexpected factors such as economic conditions, supplier disruptions, policy changes or major promotional campaigns may influence future demand. Regular updates using the latest sales data will help maintain forecast reliability.